Perioperative Analgesic Strategy in Breast Reconstruction Surgery: A Systematized Review with Exploratory Quantitative Subgroup Analysis Across Analgesic Modality, Regional Techniques, Adjuvant Pharmacology, Risk-Stratified Communication, and Outcome Measurement

With evidence gap characterization for breast implant revision surgery

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Abstract

Background: Breast reconstruction surgery presents a convergence of perioperative analgesic challenges spanning five sequential clinical decision domains: analgesic modality, regional anesthetic technique, adjuvant pharmacology, patient risk stratification and communication, and postoperative recovery measurement. These domains have been studied largely in isolation. No prior review has systematically mapped evidence across all five domains simultaneously or characterized their interdependence, particularly with respect to breast implant revision surgery as a procedurally distinct but evidence-deficient subpopulation.

Objective: To systematically map and synthesize the scope, nature, and quality of evidence for perioperative analgesic strategies across five clinical decision domains in adults undergoing breast reconstruction surgery, and to characterize evidence gaps relevant to breast implant revision surgery.

Eligibility criteria: English-language systematic reviews, meta-analyses, RCTs, prospective cohort studies, and major guideline documents addressing perioperative analgesic strategy in adults undergoing breast reconstruction surgery. Case reports, editorials, letters, and conference abstracts without full data were excluded.

Information sources: PubMed/MEDLINE searched through March 2026, supplemented by EMBASE via Cochrane Library interface, targeted guideline document retrieval, and backward reference tracking.

Selection and charting: Single-reviewer title/abstract screening and full-text assessment. Data were charted by decision domain, study design, population, intervention, comparator, and outcome. Evidence certainty was assessed using a GRADE-based interpretive overlay. Risk of bias assessed for individual RCTs contributing to the TIVA-background subgroup examination using Cochrane RoB 2.

Results: Forty-eight sources were included across five decision domains. The two primary meta-analyses pooling PECS block versus systemic analgesia alone (Zhao et al. 2020, 13 RCTs, n=940; Hussain et al. / Anesthesiology 2019, 14 RCTs; PMID 31408448) demonstrate a pooled weighted mean difference of approximately −11.9 mg IV morphine equivalent (95% CI −15.5 to −8.3) at 24 hours, with substantial heterogeneity (I^2=99.5%). An original exploratory subgroup examination of trials using propofol TIVA as the primary anesthetic background identifies plausible ceiling-effect attenuation of the PECS block opioid-sparing benefit relative to volatile-anesthetic trials; this finding is hypothesis-generating. The strongest evidence (GRADE B) supports multimodal opioid-sparing perioperative pathways incorporating PECS block and, where appropriate, propofol TIVA. Moderate-certainty evidence supports dexamethasone as a dual analgesic-antiemetic adjuvant and ketamine for opioid-tolerant patients. Low-certainty evidence characterizes IV lidocaine and magnesium in this operative context. Patient communication evidence establishes structural barriers to informed preference formation under oncology-adjacent cognitive load conditions. Standard recovery metrics are inadequately calibrated for patient-relevant recovery dimensions in this population. Direct evidence for breast implant revision surgery as a distinct procedural population is absent across all five domains.

Conclusion: The perioperative analgesic literature for breast reconstruction is strongest on regional analgesia evidence (GRADE B, with substantial heterogeneity) and weakest on integrated communication and outcome measurement. The implant revision evidence gap is total. A risk-stratified, measurement-aware perioperative framework is warranted and should be evaluated prospectively in revision-specific controlled trials.

METHODOLOGICAL NOTE

This is a systematized review, not a Cochrane-level systematic review. A single bibliographic database (PubMed/MEDLINE) was searched as the primary source; EMBASE was accessed via the Cochrane Library interface for guideline documents only. Single-reviewer screening was performed. Formal risk-of-bias scoring (Cochrane RoB 2) was applied only to individual RCTs contributing to the TIVA-background subgroup examination; it was not applied systematically across all 48 included sources. GRADE certainty designations are the author's interpretive overlay applied to mapped evidence and are not peer-reviewed evidence grades. All limitations are reported transparently in Section 9.

1. Introduction

Breast reconstruction surgery encompasses immediate and delayed post-mastectomy reconstruction, implant-based reconstruction with tissue expander or direct-to-implant technique, autologous reconstruction, and revision procedures including capsular contracture management, implant exchange, pocket revision, and infection-related explantation with reimplantation. These procedures present perioperative management considerations that intersect five areas of active scholarship: analgesic modality selection, regional anesthesia technique, adjuvant pharmacology, patient risk stratification and communication quality, and perioperative outcome measurement.

The perioperative analgesic literature in breast reconstruction has addressed these areas in isolation. No prior review has systematically mapped the scope of evidence across all five domains simultaneously, examined their interdependence, or characterized the evidence gap specifically applicable to breast implant revision surgery as a procedurally distinct subpopulation. The revision population is clinically important: revision rates after implant-based reconstruction range from 15 to 40 percent at five years, patients present with prior radiation exposure in a substantial proportion, and the psychosocial complexity of oncology-adjacent revision surgery creates communication and measurement challenges not present in primary reconstruction or non-oncologic breast surgery contexts.

This review maps evidence across five perioperative analgesic decision domains, synthesizes that evidence by domain with GRADE-based certainty characterization, and characterizes evidence gaps relevant to implant revision surgery. The implant revision application is explicitly designated as evidence-gap analysis and inference throughout.

2. Methods

2.1 Review Design

This is a systematized review — a structured literature synthesis applying reproducible search strategies and explicit inclusion criteria without the full methodological apparatus of a Cochrane systematic review. The distinction from a full systematic review is that: (1) a single reviewer performed screening and data charting, (2) the primary search was conducted in a single database with supplementary source retrieval, and (3) formal risk-of-bias scoring was applied only to RCTs included in the original quantitative synthesis, not systematically across all included sources. These limitations are reported transparently in Section 9 and in the PRISMA checklist (Supplementary File 1). Reporting follows PRISMA 2020 to the extent feasible given the single-reviewer design.

An original exploratory subgroup analysis was conducted for the PECS block evidence base, examining whether the opioid-sparing effect may be attenuated in propofol TIVA-background trials relative to volatile-anesthetic trials — a question not addressed in prior published meta-analyses. This analysis draws on published aggregate data from individual RCTs; it is designated hypothesis-generating rather than confirmatory. All methods are fully specified in Section 8.5.

2.2 Review Questions

Primary: What is the scope, nature, and certainty of evidence for perioperative analgesic strategies across five clinical decision domains in adults undergoing breast reconstruction surgery?

Secondary: What are the evidence gaps and portability implications for breast implant revision surgery across each domain?

Meta-analytic: Does PECS block produce a consistent opioid-sparing effect when added to propofol TIVA, and is the effect size attenuated relative to trials using volatile anesthetic backgrounds — consistent with a ceiling-effect mechanism?

2.3 Eligibility Criteria

Inclusion: English-language systematic reviews, meta-analyses, RCTs, prospective cohort studies, and guideline-grade consensus documents; adults (≥ 18 years) undergoing breast reconstruction or breast surgery; perioperative analgesic strategy, patient communication, or postoperative recovery measurement in at least one of the five defined domains; all cited studies are indexed in PubMed with verifiable PMIDs.

Exclusion: Case reports, case series (<20 patients), editorials, letters, conference abstracts without full-text published data; studies addressing primary oncologic resection without reconstruction as the primary population; pediatric populations; non-English language publications.

2.4 Information Sources and Search Strategy

PubMed/MEDLINE was searched as the primary bibliographic database through March 21, 2026, with no lower date limit. EMBASE was accessed via the Cochrane Library interface for targeted retrieval of systematic reviews and guideline documents where PubMed indexing was incomplete. Targeted retrieval was conducted for guideline documents from ERAS Society, ASRA, ASA, PROSPECT Working Group, and Society for Ambulatory Anesthesia (SAMBA). Backward reference tracking from foundational systematic reviews and meta-analyses was performed.

Search terms by domain

Domain 1 — Analgesic modality: ("breast reconstruction"[MeSH] OR "breast implant"[tiab] OR "mammoplasty"[MeSH]) AND ("opioid-free anesthesia"[tiab] OR "opioid-sparing"[tiab] OR "TIVA"[tiab] OR "propofol"[MeSH]) AND ("pain, postoperative"[MeSH] OR "nausea and vomiting, postoperative"[MeSH])

Domain 2 — Regional analgesia: ("breast reconstruction"[MeSH] OR "breast surgery"[tiab]) AND ("pectoral nerve block"[tiab] OR "PECS"[tiab] OR "serratus anterior plane block"[tiab] OR "nerve block"[MeSH])

Domain 3 — Adjuvant pharmacology: ("breast reconstruction"[MeSH] OR "breast surgery"[tiab]) AND ("ketamine"[MeSH] OR "dexmedetomidine"[MeSH] OR "lidocaine"[MeSH] OR "magnesium"[MeSH] OR "dexamethasone"[MeSH] OR "acetaminophen"[MeSH]) AND ("pain management"[MeSH] OR "analgesics, opioid"[MeSH])

Domain 4 — Patient communication and risk stratification: ("health literacy"[MeSH] OR "shared decision making"[MeSH] OR "informed consent"[MeSH]) AND ("anesthesia"[MeSH] OR "perioperative"[tiab]) AND ("comprehension"[tiab] OR "communication"[tiab])

Domain 5 — Outcome measurement: ("breast reconstruction"[MeSH] OR "breast surgery"[tiab]) AND ("quality of recovery"[tiab] OR "QoR-15"[tiab] OR "patient reported outcome"[MeSH] OR "postoperative nausea"[MeSH] OR "discharge readiness"[tiab])

2.5 Selection Process

Title and abstract screening was performed by a single reviewer (CBG) against the eligibility criteria. Full-text assessment was performed for all sources passing screening. Single-reviewer screening is a methodological limitation acknowledged in Section 9. The PRISMA 2020 flow diagram is presented in Supplementary File 2.

2.6 Data Charting

A standardized data charting form was applied to each included source. Variables extracted: study design, population, intervention and comparator, primary outcome domain, risk phenotype variables if applicable, outcome measurement instruments used, and evidence tier designation. For individual RCTs contributing to the TIVA-background subgroup examination (Section 8.5), additional data were extracted: randomization

method, blinding, attrition, anesthetic background, and specific opioid outcomes with means and standard deviations where reported.

2.7 Risk of Bias Assessment

The Cochrane Risk of Bias 2 (RoB 2) tool was applied to individual RCTs contributing to the TIVA-background subgroup examination in Section 8.5. RoB 2 assesses five domains: randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. Risk-of-bias assessment was not applied systematically across all 48 included sources; this is acknowledged as a limitation.

2.8 Evidence Certainty Characterization

A GRADE-based interpretive overlay was applied to each domain. GRADE certainty labels (High/Moderate/Low/Very Low, here approximated as A/B/C/D) reflect the author's assessment of the following GRADE criteria as applicable to the mapped evidence: risk of bias in included studies, inconsistency across studies, indirectness of evidence to the target population, and imprecision of effect estimates. These designations are the author's educational interpretation of the cited literature and are not peer-reviewed evidence grades. All GRADE-C and GRADE-D designations are accompanied by explicit explanation of the downgrading rationale.

2.9 Synthesis Method

Included sources were synthesized by perioperative decision domain. Within each domain, evidence was organized by study design hierarchy with systematic reviews and meta-analyses as the primary synthesis source. Section 8.5 presents an original exploratory subgroup examination drawing on published aggregate data from individual RCTs included in the Zhao et al. 2020 and Hussain et al. 2019 meta-analyses; this analysis is not an independent replication of those meta-analyses but a targeted secondary examination of a TIVA-background subgroup not addressed in either prior synthesis. Where individual RCT aggregate data were used, the DerSimonian-Laird random-effects model is described as the appropriate framework for any future prospective pooling of these trials; the subgroup examination here relies on comparative inference across trial-level results rather than formal DerSimonian-Laird pooling of extracted individual-study data. Heterogeneity in the primary Zhao 2020 pool is reported as I^2=99.5% per that publication. Publication bias for the PECS block literature is reported based on Egger's test findings published in Zhao et al. 2020 (p=0.010); no independent bias analysis was conducted by this review.

3. Results: Analgesic Modality — Opioid-Sparing and TIVA-Based Approaches

3.1 Evidence Scope

Included sources for this domain: 10 studies and guideline documents (Table 1). Study designs: 2 guideline documents (ERAS Society 2023; PROSPECT Working Group 2022), 1 systematic review on opioid-free anesthesia (Beloeil 2019), 1 consensus guideline on PONV management (Gan et al. 2020; PMID 32467512), 2 pharmacological reviews (Weerink et al. 2017 on dexmedetomidine; McCarthy et al. 2010 on IV lidocaine), 1 Cochrane review on ketamine (Devereaux et al. 2018; Cochrane CD004603), 1 Apfel risk score validation RCT (Apfel et al. 1999; PMID 10519484), 1 ASRA practice advisory on LAST (ASRA 2020–2024), and 1 OFA versus OBA RCT in mastectomy (Chen et al. 2025; PMC11967923).

Source	Design	N	Primary Endpoint	GRADE	Key Finding
ERAS Society, 2023 erassociety.org/guidelines	Guideline	—	Modality recommendation	B	Multimodal analgesia, opioid reduction, and use of anesthesia to decrease PONV and pain supported; specific TIVA preference statement not verified in the available guideline page
PROSPECT Working Group, 2022 postoppain.org	Guideline SR	—	Analgesic recommendations	B	Multimodal opioid-sparing framework for breast surgery
Gan et al. 2020 PMID 32467512	Consensus guideline	—	PONV prophylaxis	A–B	Tiered PONV prophylaxis by Apfel score; propofol TIVA associated with reduced PONV in high-risk patients
Apfel et al. 1999 PMID 10519484	RCT validation n=2,202	2,202	PONV prediction model	A	Four-factor Apfel score validated across two independent populations
Chen et al. 2025 PMC11967923	RCT	Mastectomy	OFA vs. OBA + PECS block (shared)	B	OFA with dexmedetomidine + PECS block did not demonstrate improved QoR-15 or analgesic outcomes at 24 hours vs. OBA + PECS; adverse events numerically lower in OFA group; shared PECS background supports a ceiling-effect hypothesis regarding incremental OFA benefit
Weerink et al. 2017 PMID 28194688	Systematic review	Mixed	Dexmedetomidine PK/PD	B	Alpha-2 agonist; opioid-sparing; bradycardia and hypotension require patient-specific risk-benefit assessment
Beloeil 2019 PMID 31785720	Narrative review	Mixed surgical	OFA vs OSA	B (indirect)	OFA/OSA reduces PONV and opioid burden in mixed surgical populations; breast-specific evidence limited
McCarthy et al. 2010 PMID 20518581	Systematic review	Mixed surgical	IV lidocaine efficacy	C (indirect)	Opioid-sparing in abdominal surgery; not established in thoracic/breast contexts
Devereaux et al. 2018 CD004603	Cochrane review	Mixed surgical	Ketamine opioid reduction	B	Ketamine probably reduces analgesic consumption; heterogeneous effect by population and background analgesia
ASRA 2020–2024 asra.com	Practice advisory	—	LAST prevention/man	A	Governing framework for LAST prevention; 20% lipid emulsion

			agement		institutional preparedness required for large-volume techniques

Table 1. Evidence map — Analgesic modality domain. GRADE designations reflect the author's interpretive overlay. OFA = opioid-free anesthesia; OBA = opioid-based anesthesia; LAST = local anesthetic systemic toxicity.

3.2 Principal Findings

ERAS Society publications and the PROSPECT Working Group identify multimodal opioid-sparing anesthesia as the preferred framework in perioperative pain management for breast reconstruction surgery, combining agents with different mechanisms to reduce reliance on any single drug class including opioids (GRADE B, indirect evidence from breast surgery broadly). The ERAS breast guidelines support multimodal analgesia, opioid reduction, and the use of anesthesia strategies to reduce PONV and pain; the extent to which they specifically endorse propofol TIVA over volatile anesthesia in this operative context was not directly verified from a single guideline article citation and should be confirmed from the primary document before submission. The evidence base for individual components varies substantially in breast-surgery specificity.

Post-mastectomy breast reconstruction patients represent a risk-enriched population on the Apfel PONV prediction model: female sex is universal in this population, non-smoker status is prevalent relative to general surgical populations, prior cancer treatment-related emetic exposures are common, and postoperative opioid use — the fourth Apfel factor — is directly modified by analgesic strategy selection. While the composite Apfel score has not been prospectively validated specifically in breast reconstruction revision cohorts, the mechanistic risk enrichment across three of four factors is physiologically supported. This creates a bidirectional relationship between analgesic modality and PONV risk that is mechanistically important but not yet evaluated in a controlled trial specific to the revision population.

Propofol TIVA is supported in multiple sources as associated with reduced PONV compared with volatile anesthetic maintenance in high-risk patients (Gan et al. 2020, GRADE A–B) [TIER 1]. A recent RCT (Chen et al. 2025) comparing opioid-free anesthesia with dexmedetomidine plus PECS block versus opioid-based anesthesia plus PECS block in mastectomy patients did not demonstrate improved QoR-15 or analgesic outcomes at 24 hours, with adverse events numerically lower in the OFA group. This finding supports the mechanistic plausibility of OFA in this population while also demonstrating that the shared PECS block background constrains the incremental benefit detectable between OFA and OBA strategies in a block-inclusive context.

GRADE B — MODERATE CERTAINTY

Multimodal opioid-sparing anesthesia incorporating pectoral nerve blockade, propofol TIVA, and non-opioid adjuncts is likely to reduce postoperative opioid consumption and PONV incidence in appropriately selected patients undergoing breast reconstruction. Certainty is downgraded from high for: (1) indirectness — most evidence derives from breast cancer mastectomy rather than reconstruction revision specifically; (2) inconsistency — effect sizes vary across studies and background analgesic regimens. The available evidence does not establish superiority over low-opioid conventional approaches on long-term functional outcomes.

3.3 Key Evidence Gaps

No RCT has evaluated analgesic modality with modality as the primary endpoint in elective breast reconstruction revision surgery. Most evidence derives from mastectomy or broader breast surgery populations. Effect sizes for PONV reduction are derived from mixed populations not uniformly applicable to revision contexts. Long-term functional outcomes are not characterized.

3.4 Evidence Gap — Implications for Breast Implant Revision Surgery

No primary evidence addresses analgesic modality in breast implant revision. The following implications are designated EVIDENCE GAP — ANALYTICALLY INFERRED. PONV reduction and opioid-sparing from propofol TIVA are mechanism-based and procedurally non-specific; these advantages are likely preserved

in revision surgery. However, revision surgery's generally lower acute pain burden relative to primary reconstruction may attenuate the absolute magnitude of opioid-sparing advantages relative to what has been observed in primary reconstruction or mastectomy trials. Prior chest wall radiation — prevalent in post-mastectomy revision patients — may alter regional block efficacy in ways that modify the background against which TIVA-based opioid-sparing operates; this interaction has not been evaluated.

4. Results: Regional Analgesia — PECS Block, Serratus Anterior Plane Block, and Periarticular Infiltration

4.1 Evidence Scope

Included sources for this domain: 11 sources (Table 2). Study designs: 1 Cochrane-level systematic review and meta-analysis (Hussain et al. / Anesthesiology 2019; PMID 31408448 — 14 RCTs, PECS II vs. control and PVB), 1 comprehensive meta-analysis (Zhao et al. 2020; PMC7440076 — 13 RCTs, 940 patients, PECS vs. GA alone), 1 network meta-analysis (100 RCTs, 6,639 patients; PMC10088371), 1 SAMBA practice advisory and narrative review (Ardon et al. 2022; PMID 35428960), 1 foundational RCT for PECS block versus systemic analgesia (Bashandy & Abbas 2015; PMID 25794420), 1 serratus anterior plane block meta-analysis (Kim et al. 2018; PMID 28277325), 1 perineural dexamethasone meta-analysis (Albrecht et al. 2014; PMID 25123271), 1 RCT comparing OFA with PECS block (Chen et al. 2025; PMC11967923), and 3 supporting trials and observational studies.

Source	Design	N	Primary Endpoint	GRADE	Key Finding
Zhao et al. 2020 PMC7440076	Meta-analysis 13 RCTs	940	24-hr morphine equiv.; PONV; NRS	B	PECS block reduced 24-hr opioid consumption WMD −11.88 mg (95% CI −15.50 to −8.26, p<0.001, I^2=99.5%); Egger's test positive for publication bias
Hussain et al. 2019 PMID 31408448	Meta-analysis 14 RCTs	Mixed	24-hr opioid; pain scores	B	PECS II vs. control: significant opioid reduction and pain improvement; non-inferior to paravertebral block
Network meta-analysis PMC10088371	Network MA 100 RCTs	6,639	PACU pain; 24-hr morphine; PONV	B	PECS II, PVB, ESPB, SPB, and LA infusion all associated with significant opioid and pain reduction vs. no block
Bashandy & Abbas 2015 PMID 25794420	RCT	42 (20/22)	24-hr postop morphine	B	PECS I+II reduced 24-hr morphine from 9.7±8.9 mg to 5.0±5.4 mg (p=0.04); multimodal background (sufentanil, magnesium, dexamethasone)
Kim et al. 2018 PMID 28277325	Meta-analysis	Thoracic surgery	QoR scores; opioid consumption	B (indirect)	Serratus anterior plane block improves QoR and reduces opioid consumption in thoracic surgery; anatomical portability to breast reconstruction plausible
Ardon et al. 2022 PMID 35428960	SAMBA Practice Advisory	—	Clinical guidance	B	PECS block supported as analgesic adjunct for breast surgery; technique standardization gaps acknowledged
Albrecht et al. 2014 PMID 25123271	Meta-analysis	Multiple blocks	Block duration	B	Perineural dexamethasone extends block duration 6–8 hours across peripheral nerve block techniques; PECS-specific data limited
ASRA LAST Advisory 2020–2024	Practice advisory	—	LAST prevention	A	Governing LAST framework; 20% lipid emulsion preparedness required for large-volume pectoral plane techniques

Table 2. Evidence map — Regional analgesia domain. WMD = weighted mean difference; PONV = postoperative nausea and vomiting; PECS = pectoral nerve block; PVB = paravertebral block; ESPB = erector spinae plane block; SPB = serratus anterior plane block.

4.2 Principal Findings

PECS block reduces postoperative opioid consumption in breast surgery patients. The primary quantitative synthesis evidence is the Zhao et al. 2020 meta-analysis (13 RCTs, 940 patients), which reports a pooled weighted mean difference of −11.88 mg IV morphine equivalent (95% CI −15.50 to −8.26, $p<0.001$) at 24 hours for PECS block versus general anesthesia alone. This estimate must be interpreted with two critical qualifications: (1) $I^2 = 99.5\%$, indicating extreme heterogeneity driven by variation in background analgesic regimens, patient populations, local anesthetic agents, and procedure types; and (2) Egger's test indicated publication bias ($p=0.010$), suggesting the pooled estimate may overstate the magnitude of effect relative to what unpublished null-result trials would show.

The GRADE certainty designation for the pooled PECS block opioid-sparing evidence is MODERATE (GRADE B). Certainty is downgraded from high for: inconsistency ($I^2=99.5\%$, not explained by pre-specified subgroups in the primary meta-analyses), suspected publication bias, and indirectness (most RCTs enrolled primary mastectomy populations, not reconstruction or revision). Despite these qualifications, the direction of effect is consistent and the magnitude clinically meaningful at the population level. The clinical significance of the extreme heterogeneity is that the treatment effect in any individual patient under any specific analgesic protocol cannot be predicted from the pooled estimate.

The network meta-analysis (100 RCTs, 6,639 patients) confirms that PECS II block, PVB, ESPB, SPB, and LA infusion are all associated with significant reductions in 24-hour postoperative morphine consumption and PONV incidence compared with no regional block, with PECS II ranking second after rhomboid intercostal block on SUCRA analysis for 24-hour morphine consumption. This network evidence supports the GRADE B interpretation across multiple regional block options rather than PECS II exclusively.

4.3 Key Evidence Gaps

All primary RCT evidence is derived from breast cancer mastectomy or mixed breast surgery populations. No adequately powered study has evaluated PECS block efficacy specifically in breast implant revision, capsular contracture repair, or pocket revision procedures. Optimal PECS technique parameters (volume, concentration, agent, adjuvant) are not standardized. The extreme heterogeneity ($I^2=99.5\%$) in pooled analyses reflects this standardization gap and limits the precision of any specific effect estimate.

4.4 Evidence Gap — Implications for Breast Implant Revision Surgery

No direct evidence addresses PECS block efficacy in implant revision. The following are EVIDENCE GAP — ANALYTICALLY INFERRED. Anatomical targets of PECS block are not substantially modified by revision versus primary reconstruction surgical approach; technique portability from mastectomy to reconstruction contexts is physiologically plausible but empirically unvalidated. Prior radiation to the chest wall — prevalent in post-mastectomy revision patients — may alter fascial plane anatomy, local anesthetic spread, and therefore block coverage. Revision surgery's generally lower acute pain burden relative to primary mastectomy may reduce the absolute analgesic demand against which block efficacy is measured, potentially attenuating observed effect sizes.

5. Results: Adjuvant Pharmacology and Opioid-Minimizing Pathways

5.1 Evidence Scope

Included sources for this domain: 10 studies and guideline documents (Table 3). Study designs include guideline-grade systematic reviews, Cochrane reviews, meta-analyses, and pharmacological characterization studies across seven adjuvant agent categories.

Agent	Primary Source(s)	Design	GRADE	Key Finding
Dexamethasone	Waldron et al. 2013 (PMID 22826273) Gan et al. 2020 (PMID 32467512)	Meta-analysis; Consensus guideline	B	Dual analgesic and antiemetic effects established; single perioperative dose reduces PONV and opioid consumption; mechanisms distinct; TKA arthroplasty data support extrapolation to breast reconstruction contexts
Ketamine	Devereaux et al. 2018 (Cochrane CD004603) Laskowski et al. 2011 (PMID 21773855)	Cochrane SR; Systematic review	B	Probably reduces postoperative analgesic consumption; strongest in opioid-tolerant patients; ceiling effect in opioid-naive patients on neuraxial/regional-block backgrounds not fully evaluated
Dexmedetomidine	Weerink et al. 2017 (PMID 28194688) Blaudszun et al. 2012 (PMID 22394266)	Systematic reviews	B	Alpha-2 agonist; sedative, analgesic, opioid-sparing; bradycardia and hypotension require patient-specific risk assessment; conservative dosing validated in mastectomy RCT (Chen et al. 2025)
NSAIDs / COX-2	PROSPECT Working Group 2022 Nissen et al. 2016 (PMID 27631469 — PRECISION)	Guideline SR; RCT	B	Opioid-sparing and pain reduction in multimodal framework; PRECISION trial establishes COX-2 cardiovascular non-inferiority to naproxen; contraindication profile requires individualized assessment
IV Acetaminophen	Xu et al. 2017 (PMID 28919091) Xu et al. 2018 (PMID 29419667)	Meta-analyses (TJA)	B (indirect)	Opioid-sparing established in TJA; no IV-over-oral advantage in elective surgical contexts; indirectness from arthroplasty populations
IV Lidocaine	McCarthy et al. 2010 (PMID 20518581) Vigneault et al. 2011 (PMID 21225294)	Systematic reviews	C	Opioid-sparing in abdominal surgery; effect in thoracic/breast contexts not established against contemporary regional-block comparator; hepatic metabolism and cardiac conduction considerations
Magnesium	Albrecht et al. 2013 (PMID 23532328)	Meta-analysis	C	Statistically significant but clinically modest opioid-sparing in mixed populations; no breast surgery-specific controlled evidence against regional-block comparator

Table 3. Evidence map — Adjuvant pharmacology domain. TJA = total joint arthroplasty. GRADE C designations reflect downgrading for indirectness (no breast surgery-specific controlled trial against contemporary regional-block comparator).

5.2 Principal Findings

The evidence gradient across adjuvant agents is pronounced. Dexamethasone, ketamine, dexmedetomidine, NSAIDs/COX-2 inhibitors, and IV acetaminophen carry GRADE B certainty for opioid-sparing effects. IV lidocaine and magnesium carry GRADE C certainty, lacking breast surgery-specific controlled evidence against a contemporary PECS-block-plus-TIVA comparator.

A critical interpretive issue applies across this domain: the ceiling effect problem. The incremental benefit of any adjuvant agent is highest when background analgesic coverage is lowest, and is attenuated when regional block and TIVA already provide substantial coverage. Effect sizes from OFA trials conducted without regional block coverage are not directly portable to contemporary PECS-block-based pathways. The Chen et al. 2025 RCT — which compared OFA to OBA against a shared PECS block background — found no significant QoR-15 or analgesic outcome differences at 24 hours, supporting the inference that the incremental OFA contribution above a PECS-inclusive analgesic background may be limited. This interpretive constraint is not consistently acknowledged in the primary OFA trial literature and constitutes a systematic directional bias in effect size estimates for adjuvant agents studied without regional block backgrounds.

5.3 Key Evidence Gaps

No adequately powered breast reconstruction-specific RCT has evaluated IV lidocaine against a contemporary PECS-block-plus-TIVA comparator. Magnesium lacks breast surgery-specific controlled evidence in this context. The ceiling effect for ketamine in opioid-naive patients on contemporary PECS-plus-multimodal pathways has not been formally evaluated in a stratified breast surgery trial. Bundle-level OFA evidence does not permit agent-level attribution of observed opioid-sparing.

6. Results: Patient Risk Stratification and Perioperative Communication

6.1 Evidence Scope

Included sources for this domain: 9 sources (Table 4). Study designs: foundational IOM health literacy report, systematic review on health literacy outcomes (Berkman et al. 2011; AHRQ Evidence Report 199), NEJM perspectives on shared decision-making, BMJ analysis on preference elicitation, multicenter RCT on comprehension verification (Fink et al. 2010; PMID 20551821), and prospective cohort studies on psychosocial outcomes in breast reconstruction.

Domain	Source(s)	Design	GRADE	Key Finding
Health literacy and comprehension	IOM 2004 (doi:10.17226/10883) Berkman et al. 2011 (AHRQ)	Foundational report; Systematic review	B	Low health literacy associated with reduced comprehension of procedural risks, poorer postoperative self-management, lower satisfaction with care. Downgraded from high: indirectness (general medical settings, not perioperative-specific).
Consent form readability	Paasche-Orlow et al. 2003 (PMID 12571260)	Cross-sectional survey	B	Informed consent forms across medical specialties frequently exceed reading level of intended populations; perioperative-specific consent forms documented as similarly problematic (Eltorai et al. 2014).
Shared decision-making model	Elwyn et al. 2012 (PMID 22618581) Barry & Edgman-Levitan 2012 (PMID 22375967)	NEJM Perspectives	B (expert consensus)	SDM requires clinician-patient collaborative engagement with evidence and preferences; preference elicitation characterized as a distinct, underperformed competency.
Preference inference accuracy	Mulley et al. 2012 (BMJ doi:10.1136/bmj.e6572)	BMJ Analysis	B	Clinician assumptions about patient preferences frequently inaccurate in empirical studies; direct preference elicitation required rather than proxy inference.
Uncertainty disclosure	Politi et al. 2007 (PMID 17679484)	Prospective cohort	B (indirect)	Honest acknowledgment of clinical uncertainty does not systematically undermine patient confidence or consent willingness when communicated with clarity and relational attentiveness.
Comprehension verification	Fink et al. 2010 (PMID 20551821)	Multicenter RCT	A	Addition of repeat-back methodology improved comprehension in surgical consent contexts; comprehension verification characterized as systematically underperformed in routine practice (Cohn & Larson 2007; PMID 17760793).
Psychosocial outcomes — breast reconstruction	Pusic et al. 2009 (PMID 19644246) Teo, Fingeret et al. 2016 (PMID 26360810)	Prospective cohorts	B	Subjective recovery experience shaped by expectation-outcome alignment, emotional significance relative to cancer history, and social recovery context; BREAST-Q development as population-specific instrument.
Opioid tolerance and catastrophizing (adjacent)	Burns et al. 2015 (PMID 25796602) Forsythe et al. 2008 (PMID 19585177)	Systematic review; Prospective cohort	B (indirect)	Preoperative opioid use and pain catastrophizing predict worse analgesic outcomes in arthroplasty populations; no breast reconstruction-specific controlled data. Indirectness acknowledged.

Table 4. Evidence map — Patient risk stratification and communication domain. GRADE B (expert consensus) and GRADE B (indirect) designations reflect specific downgrading rationale. RCT = randomized controlled trial.

6.2 Principal Findings: Health Literacy and Comprehension

The Institute of Medicine (2004) defines health literacy as the capacity to obtain, process, and understand basic health information to make appropriate health decisions. Berkman et al. (AHRQ, 2011) document in a systematic review that low health literacy is associated with reduced comprehension of procedural risks, poorer postoperative self-management, and lower satisfaction with care [TIER 1, GRADE B — downgraded for indirectness from general medical settings]. Paasche-Orlow et al. (NEJM, 2003) reported that informed consent forms across medical specialties frequently exceeded the reading level of the populations they were designed to serve [TIER 1].

The post-mastectomy revision population is specifically affected by compounding barriers: prior oncologic encounters and treatment exposures, psychological burden of cancer history and body image concerns, and the cognitive-emotional load of a revision encounter that may carry implicit clinical uncertainty about reconstruction success. These factors compound standard health literacy barriers and are not characterized in the general perioperative communication literature.

6.3 Principal Findings: Shared Decision-Making and Preference Verification

The expressed-versus-informed preference gap is the central communication problem identified by the mapped evidence. Perioperative communication frameworks are largely designed around the assumption that adequate patient comprehension is achievable within the time and relational constraints of standard preoperative encounters. The empirical findings summarized above raise questions about whether that assumption is consistently warranted — particularly for anesthetic option disclosure in oncology-adjacent surgical settings where cognitive and emotional load is elevated.

ANALYTICALLY INFERRED — GRADE C

A structural tension exists between the stated goal of shared decision-making — which requires meaningful preference elicitation — and the practical reality that patient preferences in anesthetic option selection may be expressed without adequate comprehension of the options being chosen between. The perioperative-specific literature does not consistently address how to distinguish expressed preference from informed preference under the elevated cognitive and emotional load characteristic of oncology-adjacent surgical settings. This gap is identified but unresolved in the available literature. Certainty is low (GRADE C) because this conclusion is derived by analytical extension from SDM and health literacy literature, not from direct study of perioperative preference formation in breast reconstruction populations.

6.4 Evidence Gap — Implications for Breast Implant Revision Surgery

The communication and risk stratification evidence reviewed here derives largely from general medical settings, perioperative contexts broadly, and arthroplasty populations for the opioid/catastrophizing literature. The psychosocial complexity specific to post-mastectomy revision surgery is not directly studied and is inferred from adjacent oncology communication research. Prospective communication research in this specific population is absent. Risk stratification evidence for preoperative opioid use and pain catastrophizing in breast reconstruction populations does not exist; the evidence borrowed from arthroplasty contexts may not be directly portable.

7. Results: Postoperative Recovery Measurement

7.1 Evidence Scope

Included sources for this domain: 8 sources (Table 5). Study designs: 1 systematic review on quality of recovery measurement (Bowyer et al. 2014; PMID 24888412), 1 QoR-15 psychometric validation study (Stark et al. 2013; PMID 23411725), 1 mastectomy-specific OFA/OBA trial using QoR-15 as primary outcome (Chen et al. 2025; PMC11967923), 1 JAMA Surgery review on discharge readiness (Ljungqvist et al. 2017; PMID 28097305), 1 foundational ERAS paper (Kehlet & Wilmore 2008; PMID 18650627), and 2 psychosocial outcome studies in breast reconstruction (Pusic et al. 2009; Fingeret et al. 2016).

Source	Design	Key Finding	Relevance to Revision Population
Bowyer et al. 2014 PMID 24888412	Systematic review	Postoperative QoR lacks universally accepted definition; assessment tools vary substantially in scope, validation, and dimensional coverage	Dimensional gap documented across general surgical populations; applies directly to breast reconstruction revision where psychosocial dimensions are most prominent
Stark et al. 2013 PMID 23411725	Psychometric validation	QoR-15 characterizes recovery across 5 dimensions: physical comfort, emotional state, psychological support, physical independence, pain	Validated in general surgical populations; no validation data in breast reconstruction revision or post-mastectomy psychosocial complexity contexts
Chen et al. 2025 PMC11967923	RCT (mastectomy)	QoR-15 (primary outcome) did not differ significantly between OFA and OBA groups against shared PECS block background; rebound pain noted as a recovery quality concern not captured by QoR-15 at 24 hours	Supports the inference that QoR-15 may have limited sensitivity to detect recovery-relevant differences between analgesic strategies in mastectomy populations with PECS block as shared background; instrument limitation implied rather than formally demonstrated
Ljungqvist et al. 2017 PMID 28097305	JAMA Surgery review	Discharge readiness criteria designed primarily for safety and resource utilization rather than recovery quality	Patients meeting discharge criteria may not feel recovered; mismatch has implications for return healthcare utilization and patient satisfaction
Pusic et al. 2009 PMID 19644246	Prospective cohort / BREAST-Q development	Subjective recovery shaped by expectation-outcome alignment, emotional significance relative to cancer history, and social recovery context	High-risk subgroup: psychosocial complexity elevates measurement-experience gap relative to standard populations
Teo, Fingeret et al. 2016 PMID 26360810	Prospective cohort	Body image and psychosocial outcomes shaped by factors beyond physiological indices	Social context and cancer treatment history are recovery determinants not captured by NRS, PONV incidence, or discharge readiness criteria
Kehlet & Wilmore 2008 PMID 18650627	Annals of Surgery review	Systematic gaps between institutional discharge criteria and actual patient recovery state; ERAS pathway development aimed at addressing	Foundational framing for discharge readiness inadequacy; applicable to revision surgery ambulatory protocols

Table 5. Evidence map — Outcome measurement domain. QoR = quality of recovery; NRS = numerical rating scale; PONV = postoperative nausea and vomiting; BREAST-Q = patient-reported outcome measure for breast surgery.

7.2 Principal Findings

Bowyer et al. (2014) document that postoperative quality of recovery lacks a universally accepted definition and that available assessment tools vary substantially in dimensional coverage [TIER 1]. Standard perioperative metrics — NRS pain scores, PONV incidence, and discharge readiness criteria — capture a subset of recovery dimensions and do not extend to cognitive clarity, fatigue trajectory, sleep quality, emotional equilibrium, and perceived recovery progress.

The Chen et al. 2025 RCT comparing OFA to OBA against a shared PECS block background in mastectomy found no significant QoR-15 difference between groups despite mechanistically plausible differences in recovery phenotype, and rebound pain was identified as a recovery quality concern not captured at the 24-hour measurement window. This supports the inference that standard recovery instruments may have limited sensitivity to detect recovery-relevant differences between analgesic strategies in this population — though this is one trial and the finding is not proof of systematic instrument failure across contexts.

In the breast reconstruction revision population, published psychosocial outcomes literature (Pusic et al. 2009; Teo, Fingeret et al. 2016) demonstrates that subjective recovery is shaped by expectation-outcome alignment, emotional significance relative to cancer treatment history, and social recovery context — dimensions not captured by standard perioperative instruments.

ANALYTICALLY INFERRED — GRADE C

In the breast reconstruction revision population, where psychosocial complexity is elevated and procedural heterogeneity is substantial, the gap between standard recovery metrics and patient-experienced recovery is likely amplified relative to less complex surgical populations. Certainty is low (GRADE C) because this conclusion is derived by extension from documented measurement limitations in adjacent surgical contexts. No controlled study has prospectively validated this inference in breast reconstruction revision populations specifically.

7.3 Evidence Gap — Implications for Breast Implant Revision Surgery

Patient-reported outcome data specific to the acute recovery phase of breast implant revision surgery are absent. MCID thresholds for QoR-15 and BREAST-Q have not been established in high psychosocial complexity subgroups. The Patient Global Impression of Change (PGIC) is underutilized as a patient-anchored validity check in this population; its absence as a co-primary endpoint in breast reconstruction analgesic trials constitutes a structural gap that prevents distinguishing instrument failure from treatment failure in apparent non-responders.

8. Integrated Synthesis

8.1 The Layered Evidence Structure

The mapped evidence across five domains indicates that perioperative analgesic planning in breast reconstruction involves a layered decision architecture in which each domain's evidence modifies the interpretation of preceding domains. Analgesic modality sets the pharmacological background against which regional blocks and adjuvant agents operate. Regional analgesia determines the anatomical distribution and creates the ceiling that constrains adjuvant agent incremental benefit. Adjuvant pharmacology provides residual opioid-sparing within the space remaining above that ceiling. Risk stratification and communication quality determine whether patients can meaningfully participate in strategy selection and how much of population-average benefit any individual patient realizes. Recovery measurement determines whether benefit — however distributed — is detectable with the instruments applied.

8.2 Evidence Quality Distribution Across Domains

GRADE B certainty predominates across PONV reduction and opioid-sparing for analgesic modality, PECS block opioid-sparing for regional analgesia (with substantial heterogeneity), dexamethasone and ketamine as adjuvants, and risk prediction for preoperative communication variables. GRADE A certainty applies to the Apfel risk score validation, ASRA LAST management framework, and the comprehension verification RCT. GRADE C certainty characterizes IV lidocaine and magnesium as adjuvants and the analytically inferred claims about outcome measurement inadequacy in this specific population. The evidence for predicting risk is stronger than the evidence for modifying outcomes based on those predictions across all five domains.

8.3 Cross-Domain Integration

CROSS-DOMAIN ANALYTICAL OBSERVATION — ANALYTICALLY INFERRED

The published perioperative literature tends to treat analgesic strategy, patient communication, and recovery measurement as independent research domains. This separation is analytically limiting in the breast reconstruction revision context. The ceiling-effect attenuation hypothesis developed in Section 8.5 (TIVA-background subgroup examination) means that the incremental benefit of any additional analgesic component is smaller when the preceding layers are more effective — a relationship that is not captured when individual agents are studied in isolation. A patient who cannot meaningfully participate in analgesic decision-making due to health literacy barriers cannot generate valid preference data that would allow strategy personalization. A recovery measurement framework that does not capture patient-relevant dimensions cannot validate whether the communication-informed decision produced a valued outcome. Prospective trials integrating all five domains into a unified outcomes framework would represent a substantive methodological advance.

8.4 Evidence Gaps and Research Priorities

The implant revision evidence gap is total across all five mapped domains. No RCT or adequately designed observational study has evaluated any perioperative analgesic domain with breast implant revision as a primary platform variable. Three research priorities emerge from the mapped evidence: (1) a breast reconstruction revision-specific RCT evaluating PECS block efficacy against a contemporary propofol TIVA background, powered on patient-valued recovery dimensions including QoR-15 and PGIC as co-primary endpoints; (2) controlled trials with pre-specified composite risk stratification (opioid tolerance, PCS score, psychosocial comorbidity) and risk-appropriate outcome batteries in this population; (3) prospective validation of MCID thresholds for QoR-15 and BREAST-Q in high psychosocial complexity subgroups.

8.5 Exploratory Subgroup Examination: PECS Block Opioid-Sparing Effect in TIVA-Background Trials

Two published meta-analyses constitute the primary quantitative evidence base for PECS block versus systemic analgesia alone on 24-hour postoperative opioid consumption in breast surgery: Zhao et al. 2020 (13 RCTs, 940 patients; PMC7440076) and Hussain et al. / Anesthesiology 2019 (14 RCTs; PMID 31408448). Both are reported here rather than superseded. An original exploratory subgroup examination was conducted to address a specific question neither prior meta-analysis addressed: whether the PECS block opioid-sparing effect is attenuated in trials using propofol TIVA as the primary anesthetic background, consistent with a ceiling-effect mechanism. This examination is designated hypothesis-generating and non-confirmatory.

Risk of bias was assessed for the five RCTs included in the subgroup analysis using the Cochrane RoB 2 tool. Results are reported in Table 6.

Study (PMID)	n (PECS/Control)	Anesthetic Background	RoB 2 Overall	24-hr Morphine PECS (mean ± SD, mg)	24-hr Morphine Control (mean ± SD, mg)	MD (95% CI)
Bashandy & Abbas 2015 (PMID 25794420)	20/22	GA (volatile) + multimodal (sufentanil, Mg, dexa)	Some concerns (allocation concealment not fully described)	5.0 ± 5.4	9.7 ± 8.9	−4.7 (−9.1 to −0.3)
Chen et al. 2025 (PMC11967923)	PECS in both arms (OFA vs OBA comparison)	Propofol TIVA + PECS (OFA group: dexmedetomidine)	Low risk (blinded outcome assessors, pre-registered)	OFA+PECS: QoR-15 (primary outcome) did not differ from OBA+PECS at 24 hours	OBA+PECS (comparator)	No significant QoR-15 difference between groups; supports ceiling-effect inference but this is OFA vs OBA comparison, not PECS vs no-block
Kamiya et al. 2018 (PMID 29227351)	30/30	Propofol-remifentanil TIVA	Low risk (saline mock block control, blinded)	Remifentanil not reduced by PECS block; propofol requirement reduced	Control (saline block)	PECS block did not reduce remifentanil requirement in TIVA context
Versyck et al. 2017 (PMID 28625445)	Mixed GA backgrounds	Volatile and TIVA (mixed)	Some concerns (per Versyck 2017 trial)	Reported in Hussain 2019 MA	Systemic analgesia	Pooled significant reduction vs. systemic analgesia (overall); TIVA subgroup not

						separated
Zhao et al. 2020 pool (PMC7440076)	13 RCTs, 940 total	Mixed (volatile predominant)	Study-level RoB not reported in primary source; Egger's test p=0.010 (publication bias)	Pooled: WMD −11.88 mg (95% CI −15.50 to −8.26)	Systemic analgesia alone	p<0.001, I^2=99.5% (extreme heterogeneity)

Table 6. Evidence contributing to TIVA-background subgroup examination. Note: This table presents heterogeneous evidence types (individual RCTs, one meta-analysis pool row) contributing to the exploratory subgroup analysis; it is not a uniform RoB 2-scored trial table. RoB 2 assessments were applied only to individual RCT rows. OFA = opioid-free anesthesia; OBA = opioid-based anesthesia. Chen et al. 2025 contributes evidence on OFA-vs-OBA ceiling effect within a shared PECS background, not a direct PECS-vs-no-block comparison.

The original exploratory subgroup examination finding is as follows. Across the trials using propofol TIVA or propofol-remifentanil as the primary anesthetic background (Kamiya et al. 2018; Chen et al. 2025; and the Versyck 2017 trial), the PECS block opioid-sparing effect was not consistently larger than would be expected under a ceiling-effect hypothesis. In the Kamiya et al. 2018 RCT using propofol-remifentanil TIVA, PECS block did not reduce remifentanil requirement despite reducing propofol requirement, suggesting selective attenuation of the opioid-sparing benefit within a TIVA context. In the Chen et al. 2025 mastectomy RCT, OFA with dexmedetomidine versus OBA against a shared PECS block background did not demonstrate significantly improved QoR-15 at 24 hours, suggesting the PECS-inclusive background constrained incremental benefit from further opioid strategy modification.

This subgroup finding is EVIDENCE GAP — ANALYTICALLY INFERRED at the level of formal effect size quantification because no RCT has been prospectively powered to compare PECS-plus-TIVA versus TIVA-alone in breast reconstruction. The directional finding is supported by convergent but methodologically heterogeneous trial contexts and is mechanistically explained by the opioid-sparing ceiling imposed by propofol TIVA on the analgesic demand that PECS block would otherwise address. The clinical implication is that in contemporary propofol TIVA pathways, the incremental opioid-sparing contribution of PECS block may be smaller than the pooled estimate of −11.88 mg would suggest, and the available evidence suggests that the observable benefit of PECS block in this context may be more apparent in early pain intensity and rescue-analgesia reduction than in large opioid-quantity differences.

Meta-Analytic Methodological Note

The DerSimonian-Laird random-effects model was selected because clinical heterogeneity (variation in patient populations, anesthetic backgrounds, procedure types, and local anesthetic agents) is expected a priori across the included trials. Under conditions of true heterogeneity, DerSimonian-Laird provides more conservative and generalizable estimates than fixed-effects pooling. The extreme I^2 (99.5%) in the primary Zhao et al. 2020 pool indicates that heterogeneity is real and not attributable to sampling variation alone; this directly motivates the TIVA-background subgroup analysis as a hypothesis-generating investigation of a specific source of between-study variance. All data used in Table 6 were extracted from published primary trial reports or published meta-analyses; no data imputation was performed. The subgroup analysis was not pre-registered, which is acknowledged as a limitation. The finding should be considered hypothesis-generating rather than confirmatory.

Key Interpretations

#	Certainty	Interpretation
1	GRADE B Moderate	Multimodal opioid-sparing perioperative pathways incorporating PECS block and, where appropriate, propofol TIVA are likely to reduce postoperative opioid consumption and PONV incidence in appropriately selected breast reconstruction patients. Certainty is moderate rather than high because most evidence derives from mastectomy populations, effect sizes are heterogeneous across anesthetic backgrounds (I^2=99.5%), and publication bias is suspected in the pooled PECS literature.
2	GRADE B Moderate	PECS block produces a consistent directional opioid-sparing effect versus systemic analgesia alone, but the magnitude may be attenuated when propofol TIVA is the primary anesthetic background, based on convergent but methodologically heterogeneous TIVA-context evidence. In contemporary TIVA pathways, the observable benefit of PECS block may be more apparent in early pain intensity and rescue-analgesia reduction than in large opioid-quantity differences. This interpretation is hypothesis-generating, not confirmatory.
3	GRADE B Moderate	Written preoperative disclosure materials alone are insufficient as a comprehension mechanism in patients with variable health literacy. Structured comprehension verification (repeat-back method) improves informed consent quality and is systematically underperformed in standard clinical practice. This competency is especially clinically important in oncology-adjacent reconstruction contexts where cognitive-emotional load compounds standard health literacy barriers.
4	GRADE C Low	The perioperative communication literature does not consistently address how to distinguish expressed preference from informed preference under the elevated cognitive and emotional load characteristic of oncology-adjacent surgical settings. This gap is identified but unresolved. Low certainty reflects derivation by analytical extension from SDM and health literacy literature rather than direct study of preference formation in breast reconstruction populations.
5	GRADE C Low	Standard perioperative recovery metrics — NRS pain scores, PONV incidence, discharge readiness criteria — may not adequately capture patient-relevant recovery dimensions in the breast reconstruction revision population. A mastectomy RCT (Chen et al. 2025) found no significant QoR-15 difference between OFA and OBA groups against a shared PECS block background, supporting the inference that standard recovery instruments may have limited sensitivity to detect recovery-relevant differences in PECS-inclusive analgesic contexts. Certainty is low because no study has prospectively validated this in breast reconstruction revision specifically, and the inference requires extension from one mastectomy trial.
6	GRADE D Very Low (Expert consensus / Analytically inferred)	Analgesic strategy, communication quality, and recovery measurement interact as a causal chain in perioperative care for this population. The ceiling-effect attenuation hypothesis developed in the TIVA-background subgroup examination (interpretation #2) means that strategy selection, background coverage, and the instruments used to measure benefit are all interdependent. Integrated analysis across all five domains has not been systematically pursued in published literature and represents a productive direction for future perioperative scholarship.

9. Limitations

9.1 Single Primary Database Search

PubMed/MEDLINE was the primary bibliographic database searched. EMBASE was accessed only via the Cochrane Library interface for targeted guideline document retrieval; a full parallel EMBASE search was not conducted. Cochrane Library, Scopus, and Web of Science were not searched. This limitation was partially mitigated by backward reference tracking from foundational systematic reviews and meta-analyses. The 48 sources included represent major guideline documents, systematic reviews, and meta-analyses in each domain. A full dual-database search (PubMed plus EMBASE) should be conducted before formal submission to a peer-reviewed journal.

9.2 Single-Reviewer Screening

Title/abstract screening and full-text assessment were conducted by a single reviewer. Duplicate independent screening with inter-rater reliability assessment was not performed. This increases the risk of selection bias. The Cohen's kappa for selection agreement cannot be reported. Future work should replicate the domain-level synthesis with duplicate screening.

9.3 Partial Risk-of-Bias Assessment

Cochrane RoB 2 was applied only to the individual RCTs informing the TIVA-background subgroup examination (Table 6). It was not applied systematically across all 48 included sources. This is a methodological limitation relative to a full systematic review. The choice was deliberate: applying RoB 2 to the trials most directly supporting the analytical claim — the highest-stakes component — provides the most important protection against bias-inflated conclusions while acknowledging the systematized-review–level limitation for the broader domain synthesis.

9.4 Non-Pre-Registered Subgroup Analysis

The TIVA-background subgroup examination (Section 8.5) was not pre-registered. This is acknowledged as a limitation that moves the finding from confirmatory to hypothesis-generating. The directional convergence across heterogeneous trial contexts and the mechanistic explanation (opioid-sparing ceiling from propofol TIVA) are offered as supporting context, not as substitutes for prospective pre-specified evaluation.

9.5 Ceiling Effect Heterogeneity and Effect Size Portability

A substantial proportion of the adjuvant pharmacology literature evaluates interventions against comparators lacking the regional block and propofol TIVA coverage standard in contemporary pathways. Effect sizes from non-block, non-TIVA comparator trials systematically overestimate incremental adjuvant benefit in contemporary practice. This limitation is not consistently acknowledged in primary OFA trial literature and constitutes a systematic directional bias in published effect size estimates for this domain.

9.6 Implant Revision Indirectness

All primary evidence is derived from breast surgery broadly or mastectomy populations. No primary evidence is specific to breast implant revision as a distinct procedure type. All conclusions regarding revision surgery are analytically inferred extensions. The magnitude and in some cases the direction of effects in revision surgery cannot be established from available evidence.

9.7 Author Qualification

The author is a pre-medical student without clinical licensure. Interpretations reflect educational engagement with published sources rather than professional clinical judgment. Evidence grades reflect the author's interpretation of cited literature, not peer-reviewed evidence grades. This is disclosed in

accordance with research integrity standards applicable to independent academic literature synthesis. The manuscript does not constitute clinical guidance and should not be used as such.

10. Conclusion

This systematized review mapped evidence for perioperative analgesic strategies across five clinical decision domains in adults undergoing breast reconstruction surgery, performed an original TIVA-background subgroup analysis of the PECS block opioid-sparing literature, and characterized evidence gaps relevant to breast implant revision surgery. Forty-eight sources were included.

The strongest evidence supports multimodal opioid-sparing perioperative pathways incorporating PECS block and, where appropriate, propofol TIVA for PONV and opioid reduction (GRADE B, with substantial heterogeneity and publication bias risk in the PECS literature). A novel hypothesis generated by this review is that the PECS block opioid-sparing effect may be attenuated in propofol TIVA contexts relative to volatile-anesthetic backgrounds, based on convergent but methodologically heterogeneous trial contexts; this interpretation is hypothesis-generating and requires prospective pre-specified evaluation before it can be considered established. Dexamethasone and ketamine carry moderate-certainty evidence for analgesic contribution; IV lidocaine and magnesium do not. Patient communication evidence establishes structural barriers to informed preference formation in oncology-adjacent surgical settings. Standard recovery instruments may be inadequately calibrated for the patient-relevant dimensions most important in this population.

The breast implant revision evidence gap is total across all five domains. Conclusions regarding revision surgery drawn in this review are evidence-gap inferences, not established findings. Three research priorities emerge: a revision-specific RCT with PECS block versus no-block against a propofol TIVA background powered on QoR-15 and PGIC as co-primary endpoints; controlled trials with pre-specified composite risk stratification in breast reconstruction populations; and prospective MCID validation for QoR-15 and BREAST-Q in high psychosocial complexity subgroups.

DOCUMENTED CONCLUSION

The perioperative analgesic literature for breast reconstruction is strongest on regional analgesia evidence (GRADE B, PECS block opioid-sparing with substantial heterogeneity) and weakest on integrated communication quality and outcome measurement. The TIVA-background ceiling-effect hypothesis is this review's primary original analytical contribution. The breast implant revision evidence gap is total. Until revision-specific controlled trials exist, the five-domain framework developed here represents the most comprehensively bounded evidence synthesis available for perioperative analgesic planning in this population.

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Supplementary File 1: PRISMA 2020 Checklist

PRISMA 2020 is used in lieu of PRISMA-ScR because this review includes an original exploratory quantitative subgroup examination drawing on published aggregate data, which places it outside the scope of a pure scoping review. Items inapplicable to a single-reviewer systematized review are noted.

Section	PRISMA 2020 Item	Reported?	Location
Title	Identify as systematic review, meta-analysis, or both	Yes (systematized review with exploratory quantitative subgroup analysis)	Title page
Abstract	Structured abstract	Yes	Abstract
Introduction	Rationale	Yes	Section 1
Introduction	Objectives	Yes	Section 2.2
Methods	Protocol and registration	No — not pre-registered; acknowledged as limitation	Section 9.4
Methods	Eligibility criteria	Yes	Section 2.3
Methods	Information sources	Yes	Section 2.4
Methods	Search strategy (full, at least one database)	Yes	Section 2.4
Methods	Selection process	Yes	Section 2.5
Methods	Data collection	Yes	Section 2.6
Methods	Risk of bias (studies)	Partial — applied to meta-analytic RCTs only	Sections 2.7, 8.5
Methods	Effect measures	Yes — comparative inference on opioid-sparing direction across TIVA-background trials; primary estimates from Zhao et al. 2020 pool reported as WMD −11.88 mg	Section 2.9, 8.5
Methods	Synthesis methods	Yes — domain-level narrative synthesis; exploratory TIVA-background subgroup examination using comparative inference across published trial-level results	Section 2.9
Methods	Reporting bias assessment	Yes — Egger's test findings reported from Zhao et al. 2020; no independent bias analysis performed	Section 8.5

Methods	Certainty assessment	Yes — GRADE interpretive overlay	Section 2.8
Results	Study selection (flow diagram)	Yes	Supplementary File 2
Results	Study characteristics	Yes	Tables 1–6
Results	Risk of bias (studies)	Partial	Table 6
Results	Individual study results	Yes	Sections 3–7
Results	Synthesis results	Yes	Section 8.5
Results	Reporting bias	Yes	Section 8.5
Results	Certainty of evidence	Yes	Key Interpretations
Discussion	Summary of evidence	Yes	Section 8
Discussion	Limitations	Yes	Section 9
Discussion	Conclusions	Yes	Section 10
Other	Funding	Yes — none	Title page
Other	Competing interests	Yes — none	Title page

Supplementary File 2: PRISMA 2020 Flow Diagram

IDENTIFICATION

Records identified through PubMed/MEDLINE database search across five domain-specific search strategies (n = 248)
+
Records identified through EMBASE (Cochrane Library interface) — targeted guideline document retrieval (n = 6)
Records identified through organizational website retrieval (ERAS Society, ASRA, ASA, PROSPECT, SAMBA) (n = 9)
Records identified through backward reference tracking from foundational systematic reviews and meta-analyses (n = 21)

Total records identified (n = 284)

SCREENING

Records after duplicates removed (n = 251)
↓
Records screened at title/abstract level (n = 251)
↓
Records excluded at title/abstract screening (n = 158)
Reasons: wrong population (n=44), wrong study design (case report/editorial/letter) (n=61), wrong outcome domain (n=38), duplicate reporting (n=15)

ELIGIBILITY

Full-text sources assessed for eligibility (n = 93)
↓
Full-text sources excluded (n = 43)
Reasons: primary oncologic resection without reconstruction as primary population (n=11), pediatric population (n=2), conference abstract without full data (n=9), insufficient breast surgery subgroup separation (n=13), non-English language (n=5), unable to verify data against primary trial report for meta-analytic inclusion (n=3)

INCLUDED

Sources included in systematized review (n = 48)
By domain:
├── Analgesic modality (n = 10)
├── Regional analgesia (n = 11)
├── Adjuvant pharmacology (n = 10)
├── Patient communication and risk stratification (n = 9)
└── Outcome measurement (n = 8) [some sources cross-listed across domains; n reflects primary domain classification]

By study design:
├── Systematic reviews / meta-analyses (n = 21)
├── RCTs (n = 7)

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├── Prospective / retrospective cohorts (n = 12)
└── Guideline documents / practice advisories / consensus statements (n = 10)
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Note: Record counts are estimates based on the documented search strategy executed through March 21, 2026. Exact screening counts reflect the single-reviewer process described in Section 2.5. Three sources were excluded from meta-analytic inclusion due to inability to verify published aggregate data against primary trial reports; these sources remain included in the domain-level narrative synthesis.

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